# *Algorithms for Greater Logistics Agility*

At the forefront of the supply chain, efficiency is not an option but a vital necessity. Our company's distribution centers operate like living organisms, with automated silos serving as their central lungs. These systems manage the temporary storage of goods that are shipped twice a week to thousands of physical stores and e-commerce platforms.

This is a massive infrastructure: each **silo** has the capacity to hold hundreds of thousands of boxes. The real technical challenge lies not only in space, but in **orchestrating the movement of boxes**. As participants in this hackathon, your mission is to design the intelligence that governs these flows. You must design the input and output algorithms to execute a perfect choreography that minimizes response times and maximizes the system's agility.

## Physical Architecture and Coordinate System

The silo is organized as a high-density three-dimensional grid. Each unit load (box) must be perfectly located so that industrial control systems can interact with it.

- Aisle: Each silo has one or more.
- Side: Each aisle has two sides with shelving (Left: 01, Right: 02).
- X-Coordinate (Length): Represents the distance from the head (entrance/exit) to the end of the aisle. Range: 000 to 999.
- Y-Coordinate (Height): The vertical level on the shelf.
- Z-Coordinate (Depth): Indicates whether the box is on the first, second, or third depth on the shelf.

Each position is defined by an 11-digit string:

01_02_003_04_01, where AISLE_SIDE_X_Y_Z

## Identification of the boxes

Each box has a unique identity via a **20-digits code**.

3010028012209309 0329

- Source: 3010028 (warehouse)
- Destination: **01220930**
- Bulk number: 90329

## Storage Logic

Boxes arrive from the sorting areas, and the system must assign them an aisle, the side, and the X, Y, Z coordinates of the location in real time.

Each box is placed in a single position.

We will create a silo with the following structure:

| Aisles | Sides | X | Y | Z |
|---|---|---|---|---|
| **4** | **2** | **60** | **8** | **2** |
| (from 1 to 4) | (1 and 2) | (from 1 to 60) | (from 1 to 8) | (1 and 2) |

The boxes will be added to the queue of the corresponding shuttle. It will not be possible to place a box at z=2 if z=1 is occupied. Similarly, we cannot retrieve a box at z=2 without first moving the box at z=1 if z=1 is occupied. If you need to get the box in z=2, you need to relocate the box in z=1 to another location.

## Movement Dynamics: The Shuttles

Shuttles are automated carts that move longitudinally along the aisles and can pick up boxes, transport them, and deposit them at various positions in the silo.

- Constraint: There is only one shuttle per height level (Y). This resource is shared: it must handle both incoming and outgoing boxes.
- Initial State: At the start of the simulation (t=0), all shuttles are at the head (x=0).

- Time Formula: The execution time of a movement depends on the travel distance of the shuttle, plus a fixed time of 10 seconds for the handling operation (either a pick or drop of a box):

$$t = 10 + d$$

*(being 10 a fixed handling time in seconds, and d is the travel distance in X).*

### Example of Cycle Calculation (pick and drop)

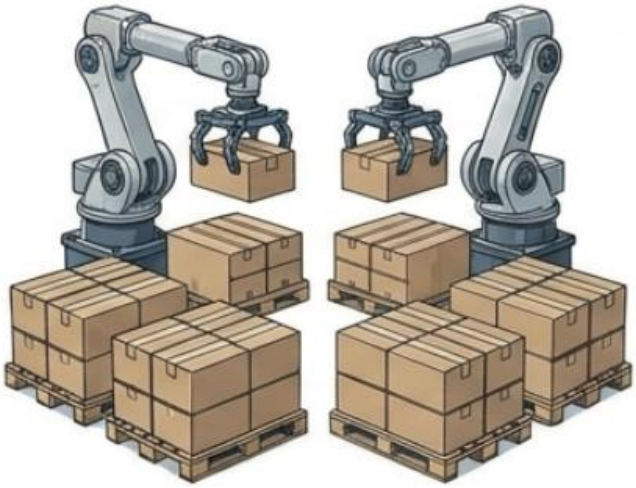
Suppose a shuttle is at X=5, a box arrives at the front, and we want to store it at x=20:

1. Trip to Head: Movement from X=5 to X=0. Time: 10 + (5-0) = 15s.
2. Trip to Destination: Movement from X=0 to X=20. Time: 10 + (20-0) = 30s.
3. Total Accumulated Time: 15 + 30 = 45 seconds

## Shipping Process and Pallet Formation

Shipping is not done randomly, box by box, but rather through the consolidation of pallets. A pallet is a set of boxes of the same destination.

- Pallet Rule: A pallet consists of 12 boxes with the same Destination.
- Palletizing: We have 2 robots that can palletize 4 pallets each at any given time, so the system can manage up to 8 reserved pallets simultaneously. Each time a pallet is shipped, we can retrieve a new one.
- Dynamic Priority: When a pallet is reserved, all its boxes are reserved at the same time. Even if Pallet 1 is reserved before Pallet 2, if we want a box from Pallet 2 to be picked first because it is in a more favorable position, the algorithm can (and should) decide to pick the box from Pallet 2 first to optimize the flow.



## The Challenge Goal

Your main objective is to develop input, output, and queue management algorithms for the shuttles that minimize total operation time.

Success Metrics

- **Full Pallets**: Percentage of pallets from which all 12 boxes are shipped.
- **Throughput** (Output Capacity): Number of pallets completed per unit of time. We can consider the average time per pallet.

Welcome to Hack the Flow.

*May your code optimize the heartbeat of our logistics!*